AMBA AHB Protocol

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Agenda

- Introduction to AHB
- AHB architecture and AHB key signals
- Bus Connections, Master & Slave Connections
- Single Transfer with wait & no wait and Multi Transfer
- Transfer Types and Transaction flow
- Burst transfers
- Differences with AXI/APB
- Use cases and References

Introduction: AHB

- AMBA — ARM's on-chip interconnect specification family
- AHB — Advanced High-performance Bus (AMBA 2/3 era)
- Designed for high-performance, high-bandwidth on-chip communication
- Used for connecting masters (CPU, DMA) to slaves (memory, peripherals)
- Designed for **high throughput**, **pipelining**, multi-master systems and burst transfers.
- Fixed bus-width (32/64/128 bits) and aligned transfers

AHB Architecture (High-level)

- Masters: initiate transfers (CPU, DMA, bus masters)
- Slaves: respond to transfers (RAM, ROM, peripherals)
- Arbiter: decides which master uses the bus
- Decoder/Interconnect: routes transactions to slaves
- HCLK/HRESET: global clock & reset signals

AHB Key Signals — Control & Data

- HADDR: Address bus
- HWDATA/HRDATA: Write / Read data buses
- HWRITE: Read(0)/Write(1) indicator
- HTRANS: Transfer type (IDLE, BUSY, NONSEQ, SEQ)
- HREADY: Transfer completion indicator
- HRESP: Transfer response (OKAY / ERROR)
- HBURST, HSIZE, HPROT: burst, size, protection info

Bus Interconnection

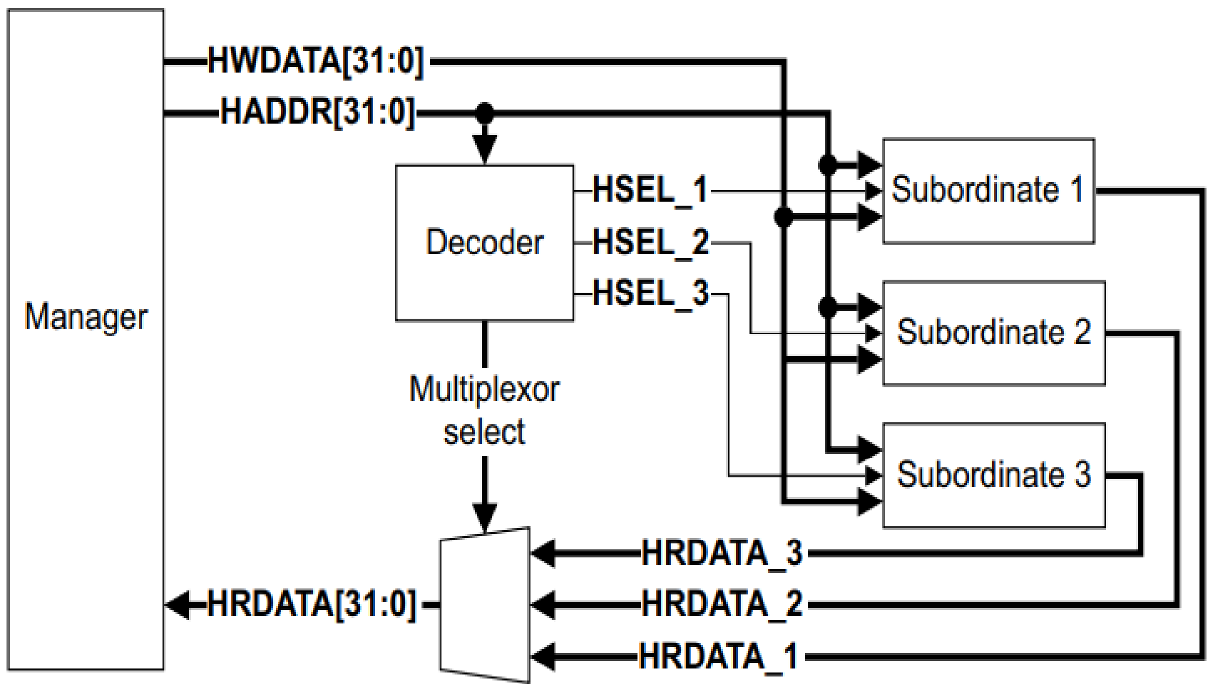


Figure 1-1 AHB block diagram

Master

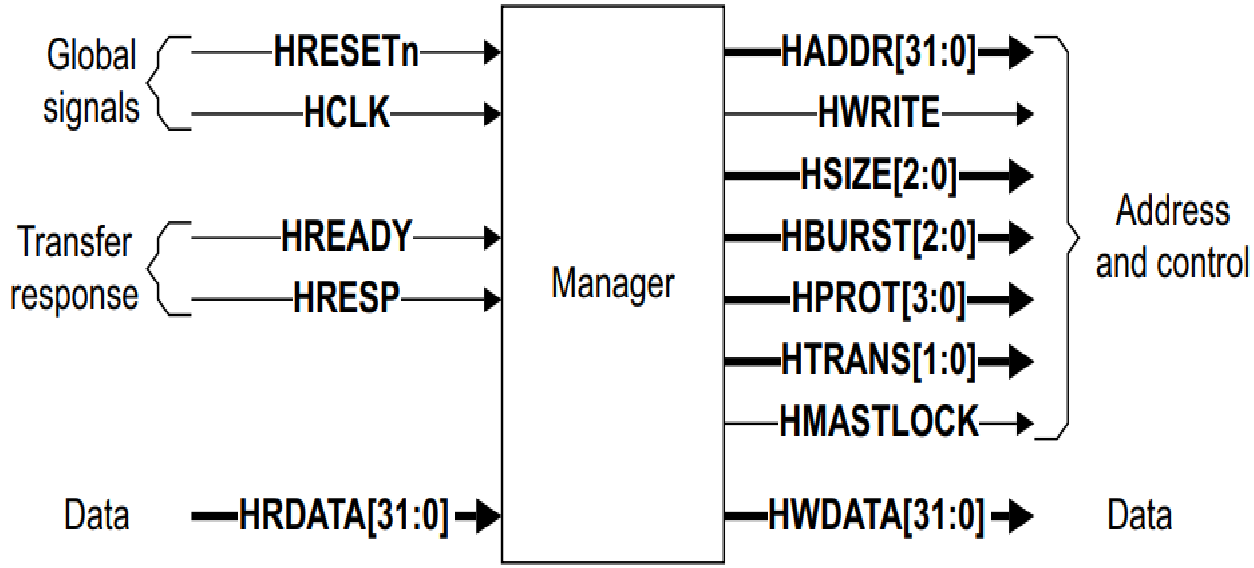


Figure 1-2 Manager interface

Slave

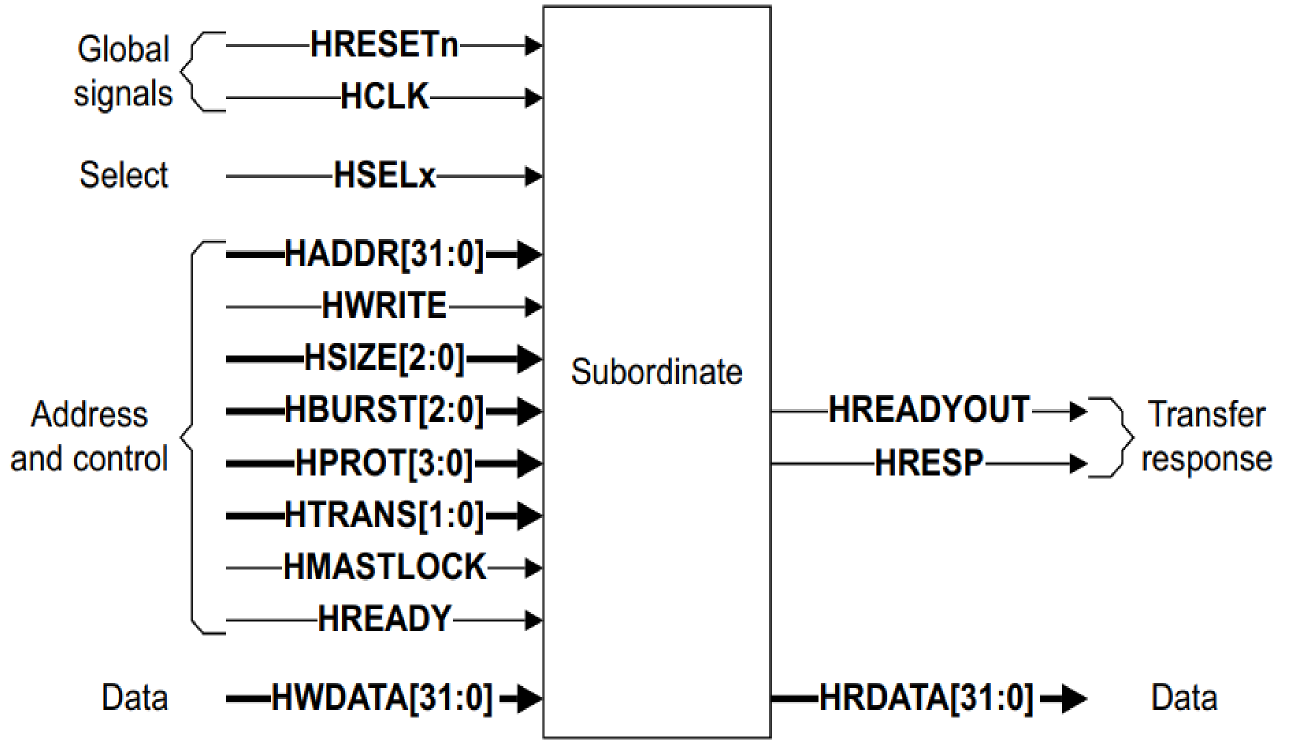


Figure 1-3 Subordinate interface

Basic Transfers: No wait

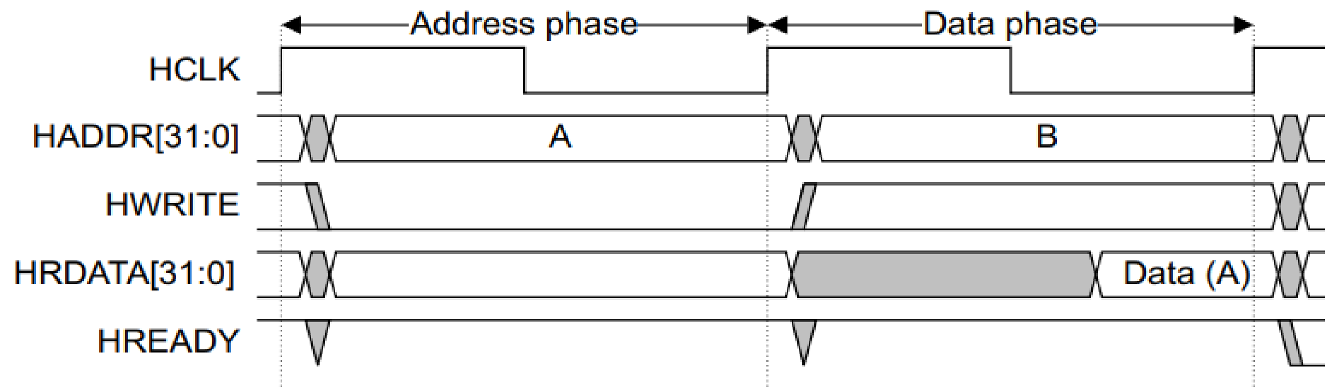


Figure 3-1 Read transfer

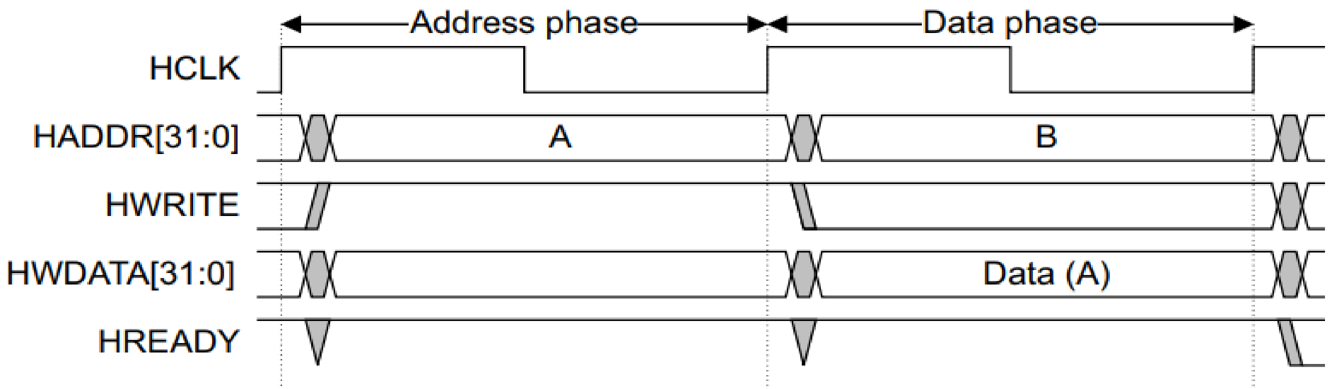


Figure 3-2 Write transfer

Single Transfer With Wait States

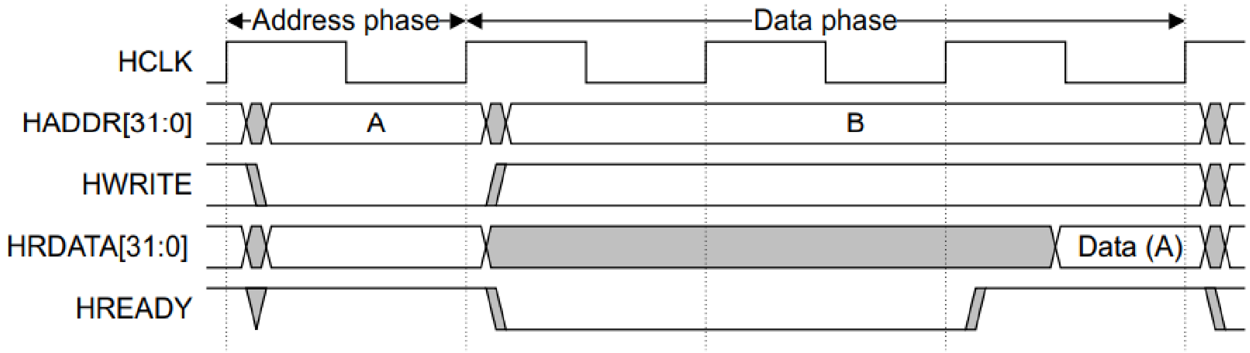


Figure 3-3 Read transfer with two wait states

Figure 3-4 shows a write transfer with one wait state.

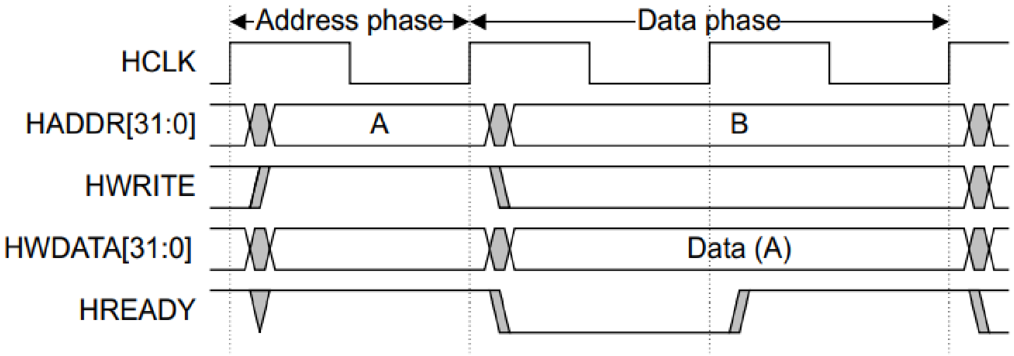


Figure 3-4 Write transfer with one wait state

Multiple Transfers

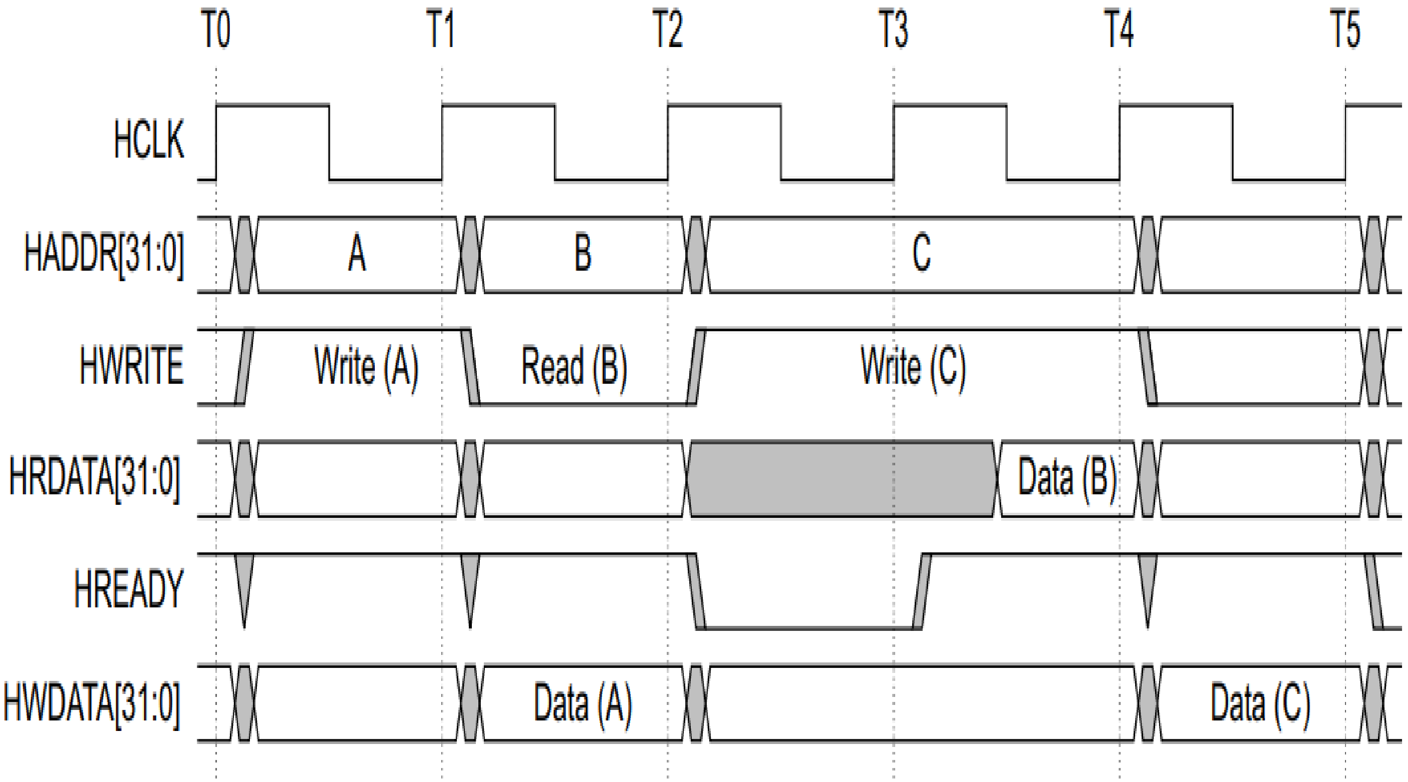


Figure 3-5 Multiple transfers

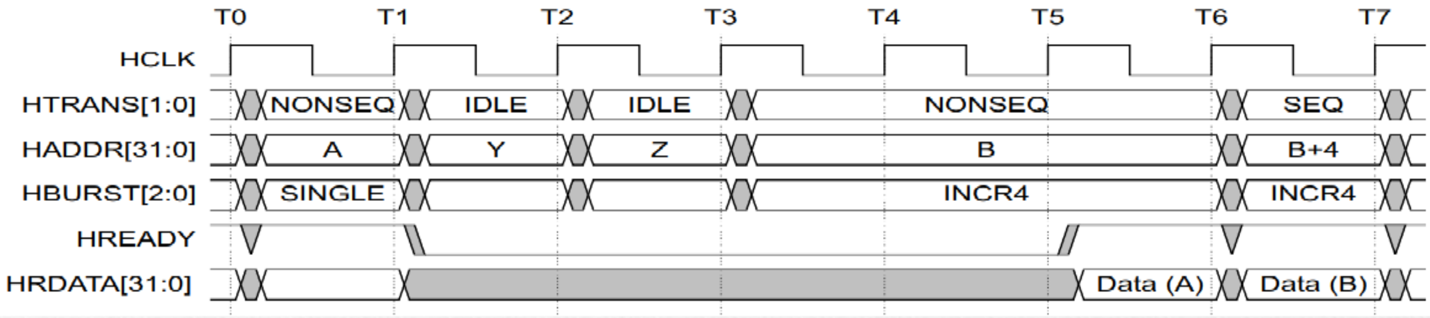
HTRANS & Transfer Types

- IDLE — no transfer
- BUSY — bus is busy (used for insertion)
- NONSEQ — start of new transfer (first in a burst)
- SEQ — sequential transfer (continuation of burst)
- Valid transfers must be NONSEQ or SEQ (not IDLE/BUSY)

IDLE transfer

During a waited transfer, the Manager is permitted to change the transfer type from IDLE to NONSEQ. When the **HTRANS** transfer type changes to NONSEQ the Manager must keep **HTRANS** constant, until **HREADY** is HIGH.

Figure 3-13 shows a waited transfer for a SINGLE burst, with a transfer type change from IDLE to NONSEQ.



BUSY transfer, fixed-length burst

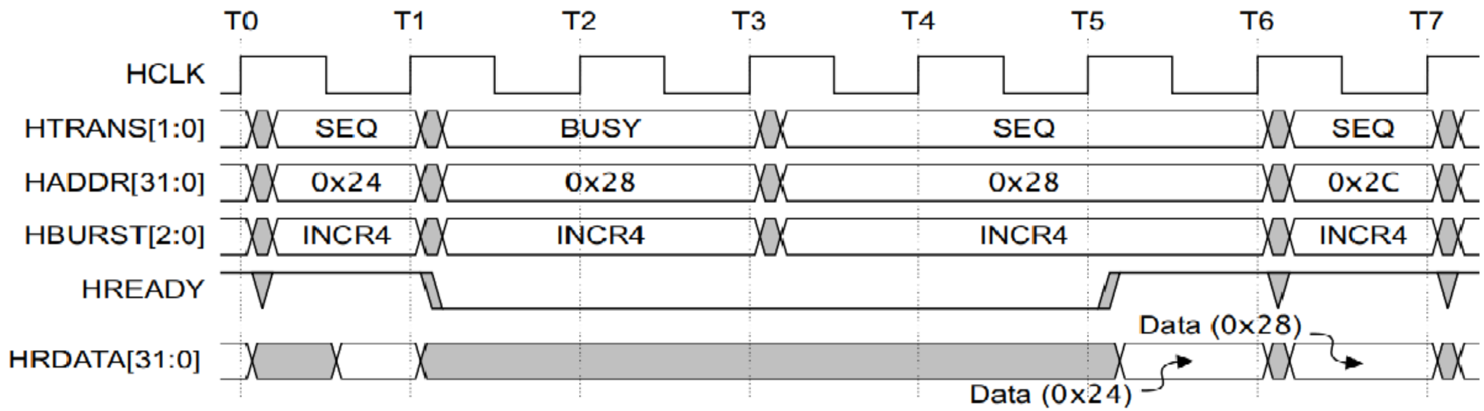


Figure 3-14 Waited transfer, BUSY to SEQ for a fixed-length burst

Transaction Flow — Pipelined Operation

- Address phase and Data phase are overlapped
- Master issues HADDR, HTRANS, HWRITE and control
- Slave responds with HREADY and HRESP in next cycles
- Pipelining increases throughput but requires careful timing
- HREADY: when low, indicates transfer in progress (stall)
- HRESP: OKAY or ERROR (single-bit or two-bit in implementations)
- Split responses allow long-latency slaves to free bus
- Masters must handle HREADY LOW and retry/continue accordingly

Burst Transfers (HBURST)

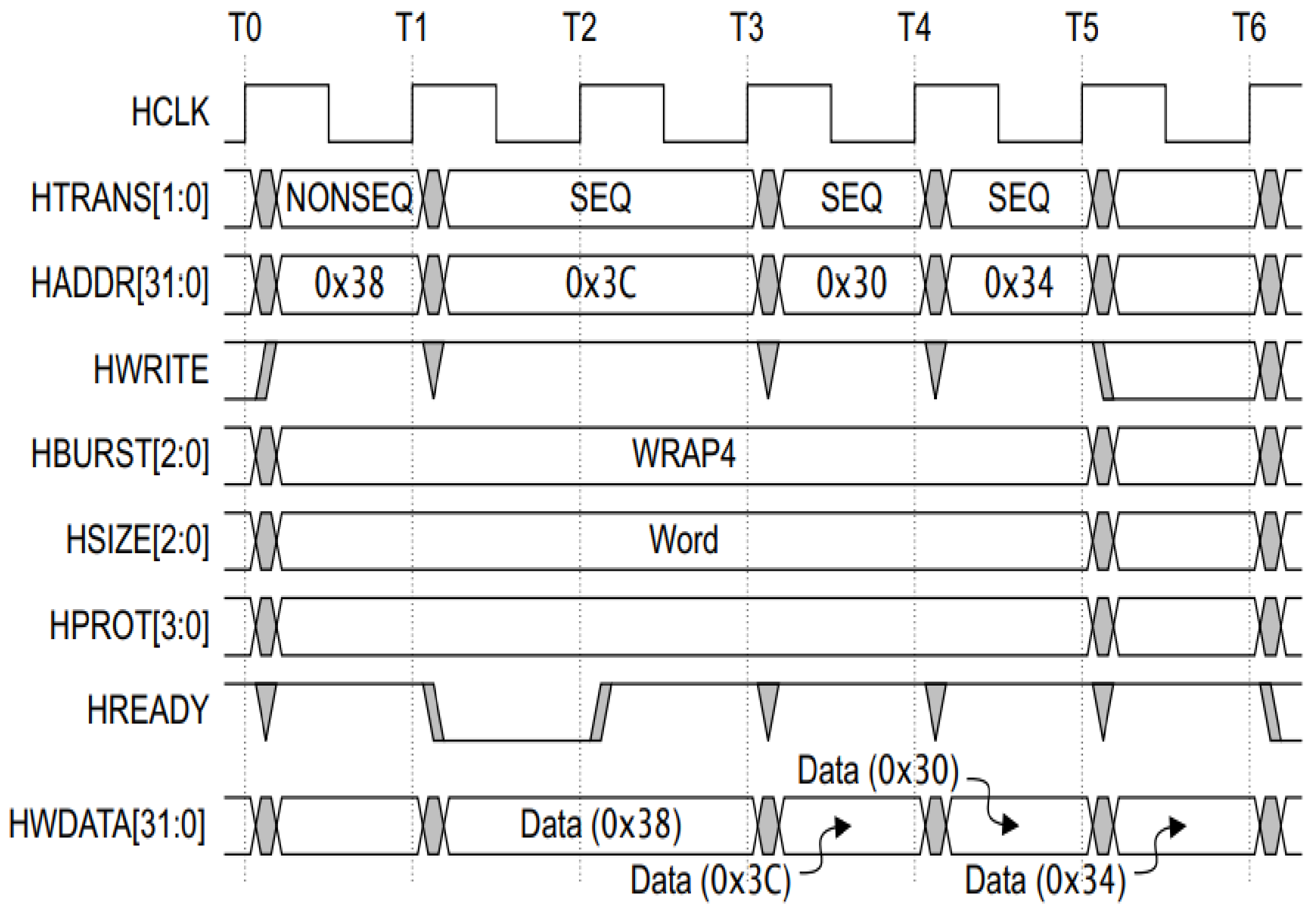
Table 3-4 Burst signal encoding

HBURST[2:0]	Type	Description
0b000	SINGLE	Single transfer burst
0b001	INCR	Incrementing burst of undefined length
0b010	WRAP4	4-beat wrapping burst
0b011	INCR4	4-beat incrementing burst
0b100	WRAP8	8-beat wrapping burst
0b101	INCR8	8-beat incrementing burst
0b110	WRAP16	16-beat wrapping burst
0b111	INCR16	16-beat incrementing burst

- Single, INCR, WRAP4/8/16, INCR4/8/16 types
- HBURST indicates how addresses progress in a burst
- WRAP bursts wrap around an aligned boundary
- INCR allows variable-length transfer sequences

Four-beat wrapping burst, WRAP4

Figure 3-8 shows a write transfer using a four-beat wrapping burst, with a wait state added for the first transfer.



Four-beat incrementing burst, INCR4

Figure 3-9 shows a read transfer using a four-beat incrementing burst, with a wait state added for the first transfer. In this case, the address does not wrap at a 16-byte boundary and the address `0x3C` is followed by a transfer to address `0x40`.

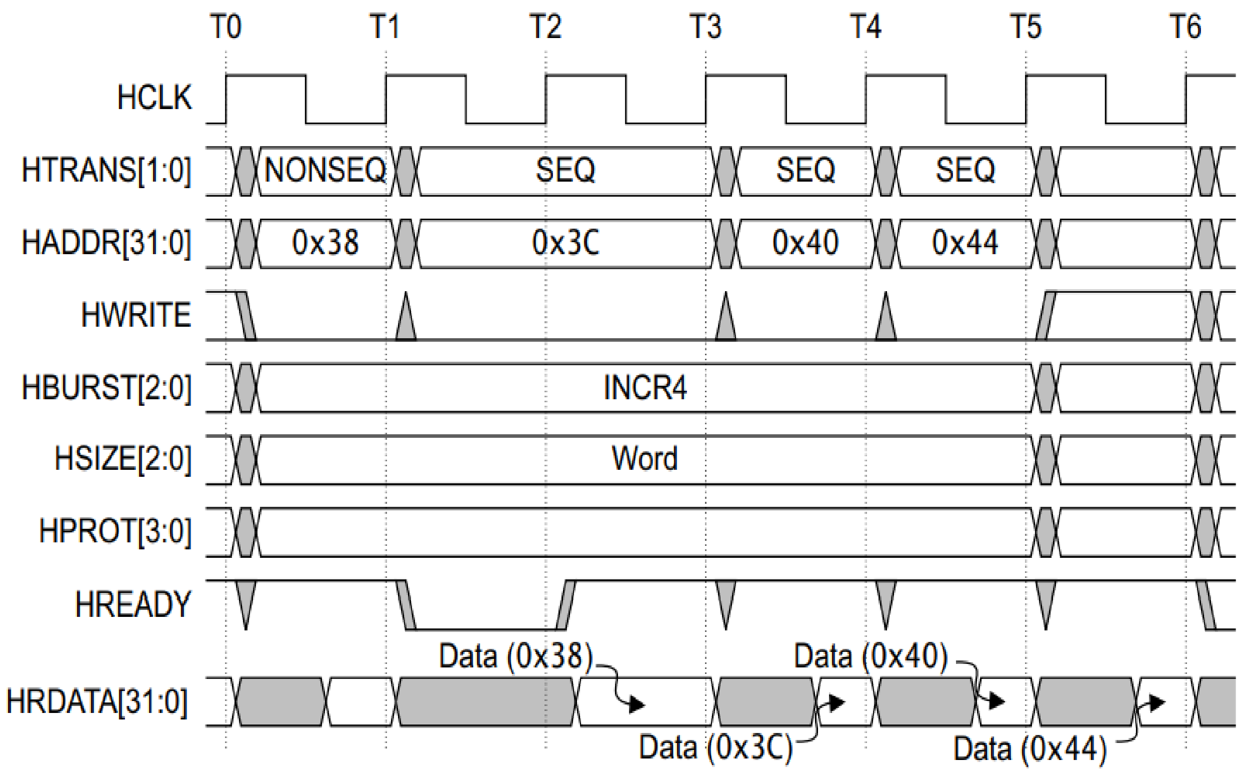
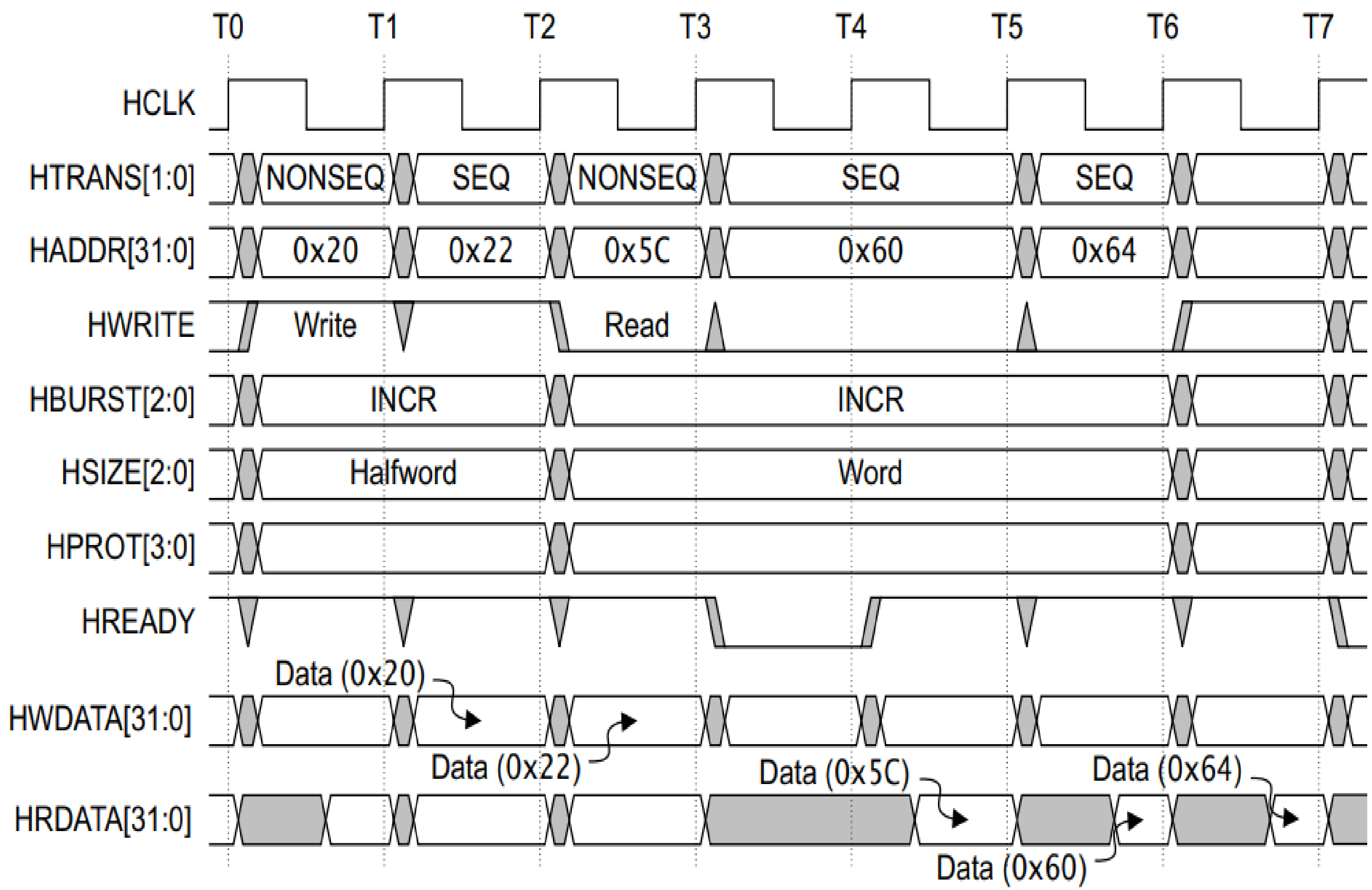


Figure 3-9 Four-beat incrementing burst

Undefined length bursts, INCR

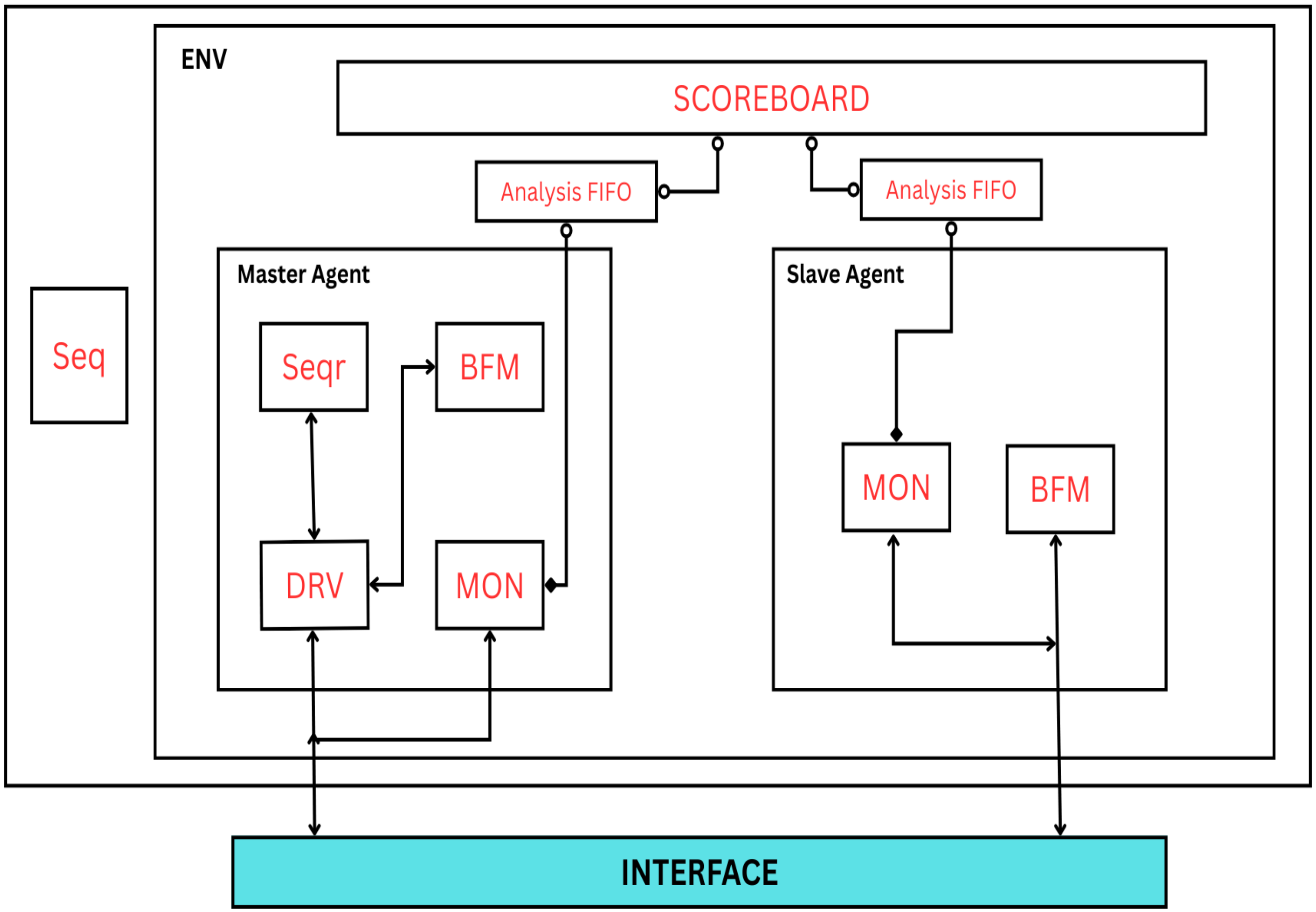
Figure 3-12 shows incrementing bursts of undefined length.



Timing Diagram — Example Read Transaction

- Cycle N: Master issues NONSEQ + address + control
- Cycle N+1: Slave samples control, master drives data (for write)
- Cycle N+2: HREADY HIGH, HRESP indicates OK/ERROR
- HREADY LOW can insert wait states, wait for HREADY HIGH then HRESP OK/ERROR is send.
- Pipelined reads: data returned one or more cycles later

AHB VIP Verification Plan



AHB vs AXI vs APB — Quick Comparison

- APB: low-bandwidth peripheral bus, simple, non-pipelined,
- AHB: pipelined, burst capable, shared bus (good mid-range)
- AXI: point-to-point, out-of-order, multiple outstanding transactions — best for high-performance

Use-cases & Real-world Examples

- On-chip DMA engines performing burst transfers to/from memory
- Microcontroller buses for connecting CPU to memory/peripherals
- Mid-sized SoCs where AXI is overkill but APB is too slow
- Interconnects in ASICs/FPGA systems using AHB-Lite modules

References & Further Reading

- ARM AMBA Specification (AHB chapter) — consult official spec for details
- UVM AHB VIP for verification examples